



DEALER BEST PRACTICE SERIES

Using Weibull To Determine Optimal Component Life Cycle Analysis

Maintenance and Repair	Application	Component Life Management	Component Renewal	MARC Management
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1.0 Introduction

Component Life Cycle Analysis (CLCA) is a process developed with the intent of analyzing component lives, with specific focus on the following objectives:

Determining realistic component replacement intervals in order to project prudent Parts Cost Variables (PCV) forecasts for commercial purposes.

To identify which components failed prematurely in order to assist in the identification of technical issues and life determining factors.

2.0 Best Practice Description

2.1 Period of analysis

Information gathered is actual component life performance data for the last three (3) years to date and / or contract term to date, with analysis frequency recorded quarterly.

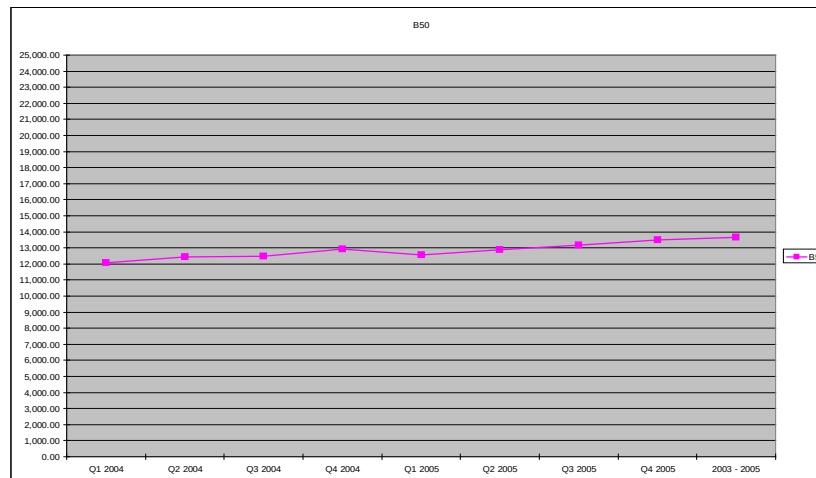
2.2 Analysis Procedure

2.2.1 Weibull analysis - (Median Rank Regression X on Y – RRY Method)

This method is used to depict the component projection life for commercial purposes with consideration to both types of data, being change-out and suspend data.

The limitation of this method is that it is not able to generate results from a small amount of data (less than 10 change-out data), In this case, use the current projection as the result.

The result from this RRY analysis will be B50 (Probability time to change-out for 50% population *See Note Below). Each result will be plotted on a trending graph on quarterly basis (See Figure 1)



(Figure 1)

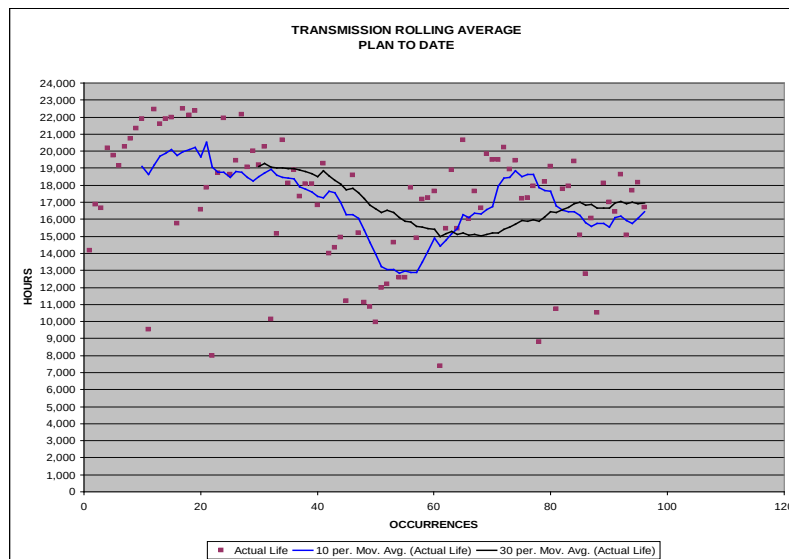
*Note: CAT Global Mining recommends B20 (Probability time to change-out for 20% population)

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2.2.2 Rolling Average

This methods calculates the average of ten (10) rolling occurrences and the average of thirty (30) rolling occurrences to identify the actual performance of the components and plot onto a trending graph. This is then used as trending and supporting data to validate and compare the projection values produced by the Weibull from the primary analysis. The source and outline criteria shall be filtered as required producing trend comparisons (see figure 2 below)

The Results for the projection life calculations use an average of the last 10 occurrences and the average of the last 30 occurrences (if applicable).



2.3 Recommendation

2.3.1 Component Interval Projection

Upon analysis of this data, the recommended projection will then be calculated using the average of all calculated results and then rounded to the next lowest multiple of 500 hours

The projection life spreadsheet will be containing the complete formula for this policy.

2.4 Occurrences Projection – Simulation

The purpose of this action is to calculate the next future occurrence that will impact from applying the current and recommended projection.

2.5 Best Performing Source

Should particular components from a particular source indicate abnormally high or low trends, option to obtain optimum component live.

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2.6 Analysis Report

The result of the above analysis will be compiled into report called “Component Life Cycle Analysis Report”.

2.7 Supporting Analysis

Depending on site capabilities and resources, each site is encouraged to generate other supporting analysis to give more realistic recommendation. The suggested analysis is: Failure mode analysis, Condition monitoring analysis, Distribution chart for yearly component life performance, and weighting analysis.

2.8 Document Structure

Each report should include:

2.8.1 Executive Summary

Summary of all analysis results and details recommended projections.

2.8.2 Raw Data

Source data sheet with date/detail of origin

2.8.3 Report section 1 – Last 3 Years data

Each major component will include the following:

Weibull with highlighted result summary.

Weibull graph

B50* quarterly trending result graph

2.8.4 Report section 2 – Contract to Date data

Each major component will include the following:

Weibull outcome with highlighted result summary

Weibull graph B50* quarterly trending result graph

2.8.5 Report section 3 – Contract to Date data

Each major component will include copies of the graph showing rolling averages and trend lines.

2.8.6 Action

Calculate for future occurrences and net risk identification.

2.8.7 Component Summary

Each major component will include:

Recommended component projection table

B50* quarterly trending result graph

Graph showing rolling averages and trend line

2.8.8 Best Performing Sources

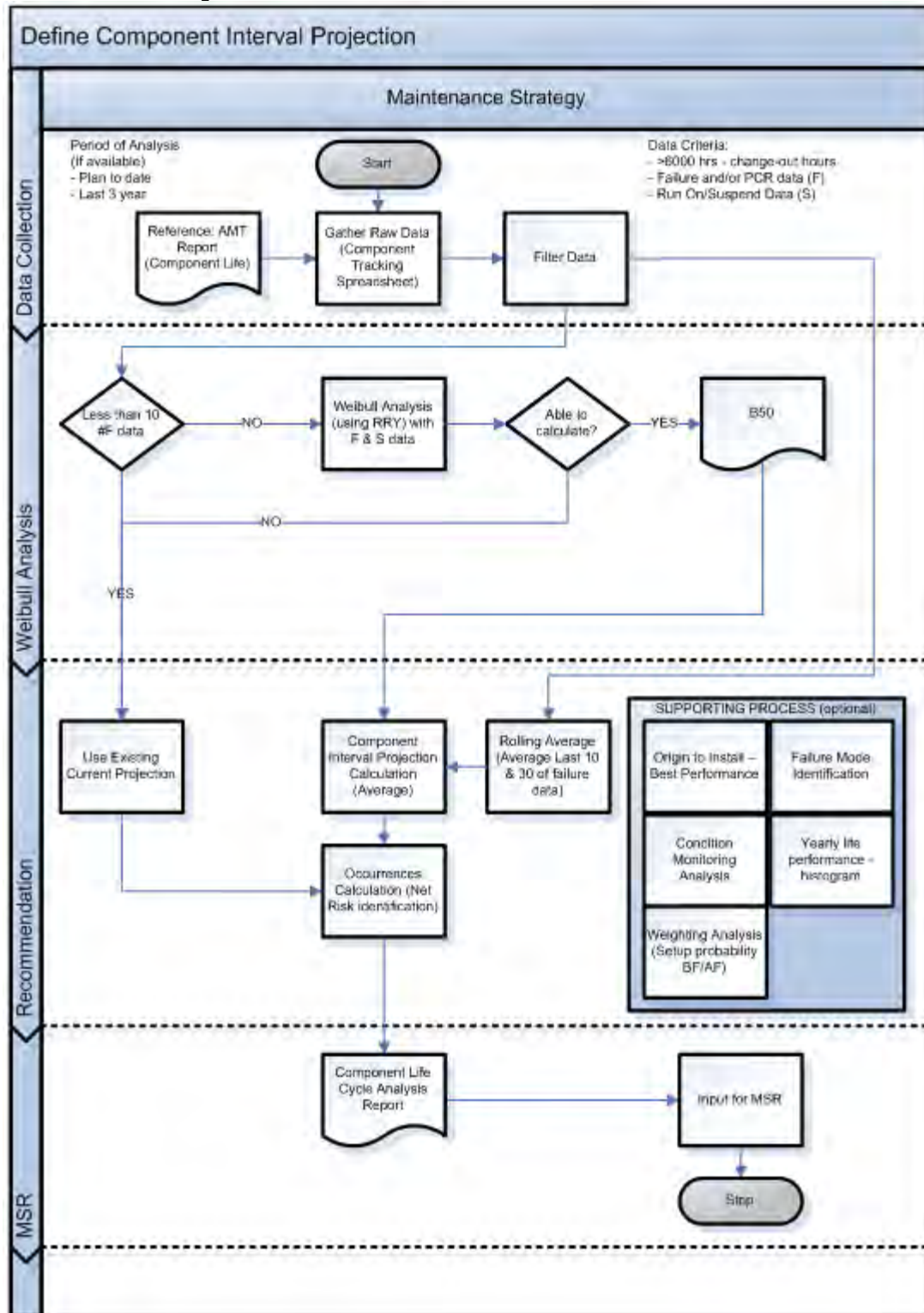
Average life of each components, source and number of events

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2.8.9 Document Formats

The report format will be distributed in PDF but the analysis will be done on an Excel spreadsheet

2.9 Process Flow and Example the Result



Current Process Flow

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COMPONENT LIFE CYCLE PROJECTION ANALYSIS Executive Summary

Site: Grasberg
 Fleet: 7938 FS
 Component Source:
 Date Issued: Q3 Calculation
 Prepared By:

COMPONENT	QUOM	Change-out Principle	Current Projection Life	Weibull RRX B50 (Last 3 years)	Weibull RRX B50 (Contract To Date)	Rolling Average (Last 10)	Rolling Average (Last 30)	Projection Recommendation
1000-Engine	Hours	Fixed	14,500	14,007	0	14,182	14,512	14,000
3030 - TRANSMISSION	Hours	0	14,000	12,595	0	12,127	12,725	12,000
3101 - TORQUE CONVERTER	Hours	0	16,000	14,236	0	16,305	14,655	15,000
3258 - DIFFERENTIAL	Hours	0	21,000	21,156	0	16,604	14,930	17,500
4006 - FRONT WHEEL GP	Hours	0	15,000	14,410	0	15,153	15,398	14,500
4008 - REAR WHEEL GP	Hours	0	14,000	12,956	0	13,613	13,318	13,000
4303 - STEERING CYLINDER	Hours	0	28,000	13,831	0	14,436	15,114	14,000
5102 - HOIST CYLINDER	Hours	0	28,000	53,700	0	12,313	12,313	26,000
7201 - FRONT SUSPENSION	Hours	0	11,000	11,273	0	12,043	11,440	11,500
7213 - REAR SUSPENSION	Hours	0	11,000	12,017	0	13,769	13,519	13,000

Example of result calculation

3.0 Implementation Steps

3.1 CLCA

CLCA is a periodic analysis, requiring accurate data for use as a reference for future strategy implementation and financial reporting. Therefore it may be necessary that the raw data supplied should be validated from site management prior to distribution.

3.2 Data Cleansing

The initial data cleansing (removal of data where double ups of Work Order (WO) number & other common data errors), and loading of accurate component history is a one-time exercise, but is absolutely essential. The amount of labor required to cleanse (remove errors) from the data, is dependent on the number of components being tracked and the quality of component history records itself.

3.3 Maintenance Strategy Reference (MSR)

In conjunction with the MSR, the result of this analysis will be used as input for next MSR review. AMT life cycle projection hours will be updated once a year in conjunction with the budget & MSR (If the quarterly data shows an exceptional differential then the forecast may be reviewed immediately in a controlled manner if authorized by the relevant manager)

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4.0 Benefits

4.1 **Prudent Parts Cost Variable (PCV) Forecast**

Availability of a standard process used to determine component change out intervals

Ability to identify occurrences and financial impact on component change out interval

4.2 **Component early failure indicators**

This factor can be analyzed utilizing rolling average data and beta # (trend) of Weibull analysis

4.3 **Maintenance Practice Indicators**

This factor can be analyzed utilizing rolling average data and beta # (trend) of Weibull analysis

4.4 **Application and Operations Indicators**

This factor can be analyzed utilizing rolling average data and beta # (trend) of Weibull analysis

4.5 **Strategy Input**

By CLCA report information, it will help equipment management planning across sites to allow improved decision making in terms of strategy improvement (i.e. Interval replacement changes, repair option selection and also component's source)

5.0 Resources Required

The main engine of this analysis is the understanding of using Weibull statistics method and able to used as Life Data Analysis Tools. For this reason, it is important to install related application as our main tools to produce analysis (i.e. Reliasoft Weibull++, Minitab, etc). Joint the training session from software vendor it is gives more values for Component Analyst to enhance his/her ability to optimize Weibull analysis on Reliability engineering.

The report is produced using Microsoft Excel, with development looking to be completed to make all calculations automatically integrated within available Weibull software.

The Main source of raw data is in AMT system, and there are web-based tools to import data based on specific filter and cleansing, thus reducing errors caused by human data entry.

6.0 Supporting Attachments / References

CLCA Intro.pdf

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7.0 Related Best Practices

0906-5.05-1009 - *MARC Rate Development*

0906-3.1-1020 - *MCTS Used to Monitor & Manage Component Life*

1106-4.5-1044 - *Utilizing Component Life as a Valuable CRC KPI*

8.0 Acknowledgements

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